



Report

Robot Description and URDF

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1. Pre-requisites

- Ubuntu 24.04 Noble runs on your computer (VM, UTM, or dual boot).
- ROS2 Jazzy is installed.
- This section is in simulation only. The hardware is not required.

2. Tasks

This first package introduces to urdf: this is the description of your robot in the ROS2 environment in a xml format. It includes gazebo, a physics engine, and rviz, a topic visualizer.

Tasks	Code
Install gazebo	<code>sudo apt-get install ros-jazzy-ros-gz</code>
Install rviz2	normally, rviz2 is already installed with ROS2 jazzy desktop
Clone the repo from GitHub	<code>git clone https://github.com/ROS2-rmitbot/lesson1_ws.git</code>
Navigate to the workspace	<code>cd ~/lesson1_ws</code>
Remove the folders build install log	<code>rm -rf build install log</code>
Build the packages	<code>colcon build</code>
Source the workspace	<code>source install/setup.bash</code>
Open VSCode	<code>code .</code>
In a terminal, launch rviz and gazebo	<code>ros2 launch rmitbot_description display.launch.py</code> <code>ros2 launch rmitbot_description gazebo.launch.py</code>
Alternatively, launch the bringup file	<code>ros2 launch rmitbot_bringup rmitbot.launch.py</code>
Look at the topic list	<code>ros2 topic list</code>
Look at the node list	<code>ros2 node list</code>
Look at the node/topic graph	<code>ros2 run rqt_graph rqt_graph</code>

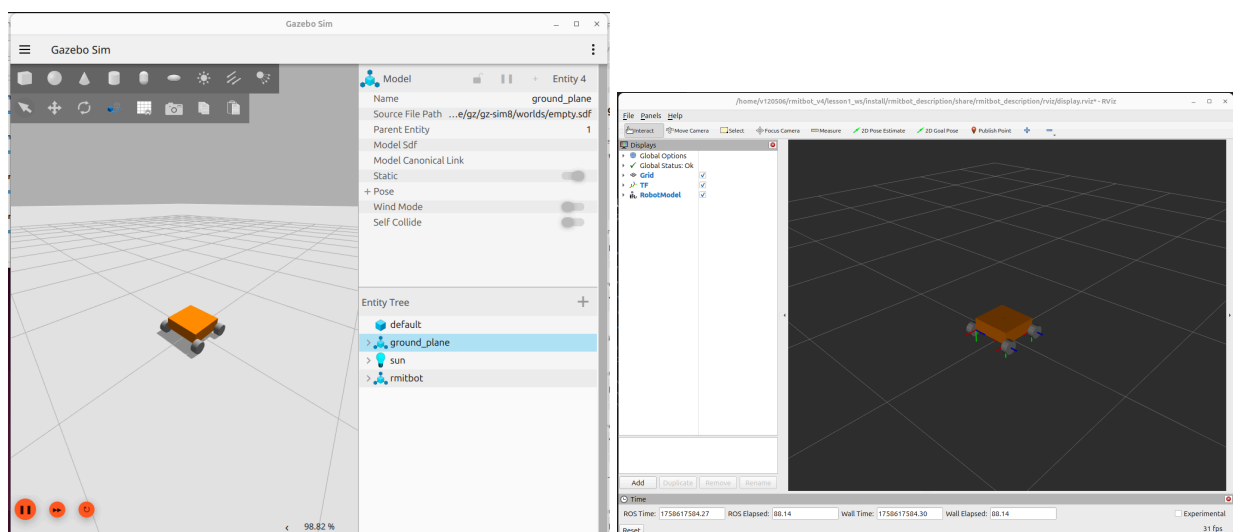


Figure 1: Gazebo is useful when the hardware is not available. Rviz2 is a graphical interface to visualize topics.

3. Explanations

This may be the first time you are using Ubuntu and ros2. The learning curve is steep, but rewarding in the long run. General knowledge about ubuntu and ros2 are:

- sudo: it means “superuser do ...”. Namely, launch a program with admin rights.
- ros2 launch ...: it means “in the ros2 framework, launch a specific program”.
- Students are invited to use resources to have an understanding about the provided code.

ROS2 stands for “Robotic Operating System”, which can be confusing as it does not look like Windows/MacOS. However, Windows loads peripherals like the mouse, the keyboard, the computer screen, the microphone, the webcam, etc. Similarly, ROS2 loads the peripherals of the robot, e.g., the motors, the Lidar, the camera, etc. Students are invited to read more about the ROS2 framework.

This package is called rmitbot_description. It describes the robot and its environment. The bringup launch file starts the following packages:

- Display: this starts Rviz
- Gazebo: this is the physics engine of ROS2.

The robot can be visualized in Rviz and gazebo, but it cannot move. This is covered in the next package rmitbot_controller. Please look at the code located in the src folder to gain insight about the ROS2 folder architecture.

- The folder **launch** contains the launch file.
- The folder **meshes** contains stl files.
- The folder **rviz** contains a configuration file for the rviz node.
- The folder **urdf** contains the description of the robot in a xml format.
- The **CMakeLists.txt** is used for compilation of the package. The folder launch, urdf, rviz, meshes must be added to the path for compilation.
- The **package.xml** is used for compilation of the package and must contain the executable dependencies: robot_state_publisher, joint_state_publisher_gui, rviz, ros2launch, ros_gz_sim for gazebo.

Rviz and gazebo looks like similar tools, but they have different purposes:

- Rviz is just for visualization: it subscribes to topics, such as robot frames (TF), and later on sensors can be visualized (IMU, Lidar, encoders).
- Gazebo is for simulation: Usually, it is used at the early stage to make sure that all the nodes for communication are working, then the code can be deployed to the real robot. The encoder, the imu, the lidar, and even the camera and depth camera can be simulated in gazebo through plugins.

4. Packages, nodes, and topics

Table 1: packages of the lesson 1

Packages	Nodes	Description
rmitbot_bringup	n/a	Launch the other packages and nodes
rmitbot_description	display.launch.py gazebo.launch.py	Launch the urdf and static tf

Table 2: nodes and topics of the lesson 1

Package/node	Topic	Description
ROS2	/parameter_events	n/a
	/rosout	n/a
rviz2	/clicked_point	Click point in rviz
	/goal_pose	Target pose of the robot (for nav2)
	/initialpose	Initial pose of the robot (for slam_toolbox)
gazebo	/clock	Sim time for node synchronization
joint_state_publisher_gui	/joint_states	Position, velocity, effort of robot joints. Normally, it is given by the robot's hardware (encoder), but for the moment, joint_state_publisher is used to fake it.
robot_state_publisher	/robot_description	Description of the robot in xml format (also called urdf)
	/tf	Dynamic tf (odom --> base_footprint)
	/tf_static	Static tf (base_footprint --> rest of the robot)

5. Future works

The urdf of the robot is quite simple: it is based on cubes and cylinders. One of the tasks is to model your robot on CAD software and transfer it to urdf format.